

Welfare Economics

Ques-01:- Define social welfare. Critically explain the following criteria as a measure of social welfare:

- (i) Growth of GNP
- (ii) Bentham's Criteria
- (iii) Cardinalist approach

Answer:-

Social Welfare:-

The objective of welfare economics is the evaluation of the social desirability of alternative economic states. An economic state is a particular arrangement of economic activities and of the resources of the economy. Each state is characterized by a different allocation of resources and different distributions of the rewards for economic activity. Although the economist may not always be able to prescribe a method by which one state of the economy can be transformed into another, policy measures frequently will be able for changing an existing situation. It is important to know in such cases whether the contemplated change is desirable.

For example, let that the economy can

attain multimarket equilibrium at two different sets of commodity and factor prices. Since the desires of consumers and entrepreneurs are consistent at both equilibria, society can choose between them, on welfare grounds. The principles by which such problems might be solved fall within the domain of the welfare economics.

The welfare of a society depends, in the broadest sense, upon the satisfaction levels of all its consumers. But almost every alternative to be judged by welfare economists will have favorable effects on some people and unfavorable effects on others. In light of this, the economist has two choices. (i) He may decline to deal with cases in which a proposed social change improves the lot of some and deteriorates the lot of others and content himself with analyzing situations in which unambiguous welfare improvements are possible. (ii) Alternatively, he may decide to make interpersonal comparisons of utility and analyze a broader class of situations. In the former case he is primarily concerned with efficient allocation of resources. In the latter, he must make explicit value judgements.

In principle one would hope that these will rest on a social consensus, since the economist has no more competence than anyone else to say that a particular move is desirable if it has unfavorable effects upon some members of society. Thus, welfare economics is concerned with the evaluation of alternative economic situations from the point of view of the society's well-being.

To evaluate alternative economic situations we need some criterion of social well-being or welfare. The measurement of social welfare requires some ethical standard and interpersonal comparisons, both of which involve subjective value judgements. Objective comparisons and judgements of the deservingness or worthiness of different individuals are virtually impossible. Various criteria of social welfare have been suggested by economists at different times.

(i) Growth of GNP as A Criterion of Welfare:-

Adam Smith implicitly accepted the growth of the wealth of society, that is, the growth of the gross national product, as a welfare criterion. He believed that

~~economic growth resulted in the increase of social welfare because growth increased employment and the goods available for consumption to the community. To Adam Smith, economic growth meant bringing W closer to W^* .~~

Assume that the total welfare in a country is W , but given the factor endowments and the state of technology, suppose that this welfare could be larger, for example W^* . The tasks of welfare economics are:

- (i) to show that in the present state $W < W^*$
- (ii) to suggest ways of raising W to W^*

(ii) Growth of GNP as a Criterion of Welfare:-

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The growth criterion implies acceptance of the status quo of income distribution as 'ethical' or 'just'. Furthermore, growth may lead to a reduction in social welfare, depending on who avails mostly from it. However, the growth criterion highlights the importance of efficiency in social welfare. Given that social welfare depends on the amount of goods and services (as well as on their distribution) efficiency is a necessary prerequisite for the maximisation of the level of welfare.

We note here that economic efficiency can be defined objectively, and the modern welfare economics is mainly concerned with the examination (comparison) of the (Pareto) - efficiency of different economic situations. However, efficiency, although a necessary condition, is not sufficient to guarantee the maximisation of social welfare. Efficiency does not dispose of the need of having an ethical standard of comparing alternative economic states.

(ii) Bentham's Criterion:-

Jeremy Bentham, an English economist, argued that welfare is improved when 'the greatest good (is secured) for the greatest number'. Implicit in

this dictum is the assumption that the total welfare is the sum of the utilities of the individuals of the society. The application of Bentham's ethical system to economics has serious short-comings. To illustrate the pitfalls in Bentham's criterion let us assume that the society consists of the three individuals, A, B, and C, so that

$$W = U_A + U_B + U_C$$

According to Bentham $\Delta W > 0$ if $(\Delta U_A + \Delta U_B + \Delta U_C) > 0$. However assume that the change which resulted in the changes in the individual utilities is such, that A's and B's utility increases, while C's utility decreases, but $(\Delta U_A + \Delta U_B) > |\Delta U_C|$. In other words, two individuals are better-off while the third is worse-off after the change has taken place, but the sum of the increases in utilities of A and B is greater than the decrease in the utility of C.

Bentham's criterion, obviously, implies that A and B have a greater 'worthiness' than C. That is, implicit in Bentham's criterion is an interpersonal comparison of the deservings of the members of the society. There is another difficulty with the application of Bentham's criterion.

This criterion cannot be applied to compare situations

one where 'the greatest good' and 'the greatest numbers' do not exist simultaneously. For example assume that in a situation $U_A = 200$, $U_B = 50$, $U_C = 30$, so the total utility in the society is 280. In another situation assume that a change occurred and $U_A = 100$, $U_B = 80$, and $U_C = 80$, so that the total utility is 260. The first situation has 'the greatest good' ($280 > 260$), but the second involves a more even distribution (of a smaller 'total good') among the 'greatest numbers'.

(iii) Cardinalist Approach:-

Several economists proposed the use of the 'law of diminishing marginal utility' as a criterion of welfare. Their argument can be illustrated by the following example. Assume that the society consists of three individuals; A has an income of £1000, while B and C have an income of £500 each. Consumer A can buy double quantities of goods as compared to B and C.

However, given the law of diminishing marginal utility, A's total utility is less than double the total utility of either B or C. Thus, $U < U^*$. To increase social welfare income should be redistributed among the three individuals. In fact cardinal welfare theorists would

maintain that social welfare would be maximised if income was equally distributed to all members of the society.

The cardinalist approach to welfare has a serious flaw: it assumes that all individuals have identical utility functions for money, so that with an equal income distribution all would have the same marginal utility of money. This assumption is too strong. Individuals differ in their attitudes towards money. A rich person may have a utility for money function that lies far above the utility (for money) function of poorer individuals. In this case a redistribution of income (towards more equality) might reduce total welfare. Opponents of the cardinalist approach pointed out also that welfare effects on equal distribution of income cannot be examined in isolation from the effects on resource allocation (which would follow the redistribution of income) and the incentives for work of the various individuals.

An equal income distribution may induce some people to work less, thus leading to a reduction in total GNP. Similarly, equal incomes in all employments may lead to an allocation of resources which produces a smaller

total output. In both cases, income equality results in (Pareto) inefficiency in the use of resources and a reduction in social welfare.

Ques-03:- Explain the underlying assumption about the marginal utility of money for being the Hicks-Kaldor compensation be correct.

Answer:-

The Kaldor-Hicks Compensation Principle:-

Nicholas Kaldor and J.R. Hicks put forward the welfare criterion based upon the compensating payments in 1939. If a certain change in economic organization or policy, according to Kaldor, makes some people better off and the others worse off, there will be a net increase in social welfare, when the gainers in welfare compensate the losers ~~and~~ and are still better off than before.

Assumptions:-

(i) An individual himself is the best judge of his satisfaction which is independent of the satisfaction of others.

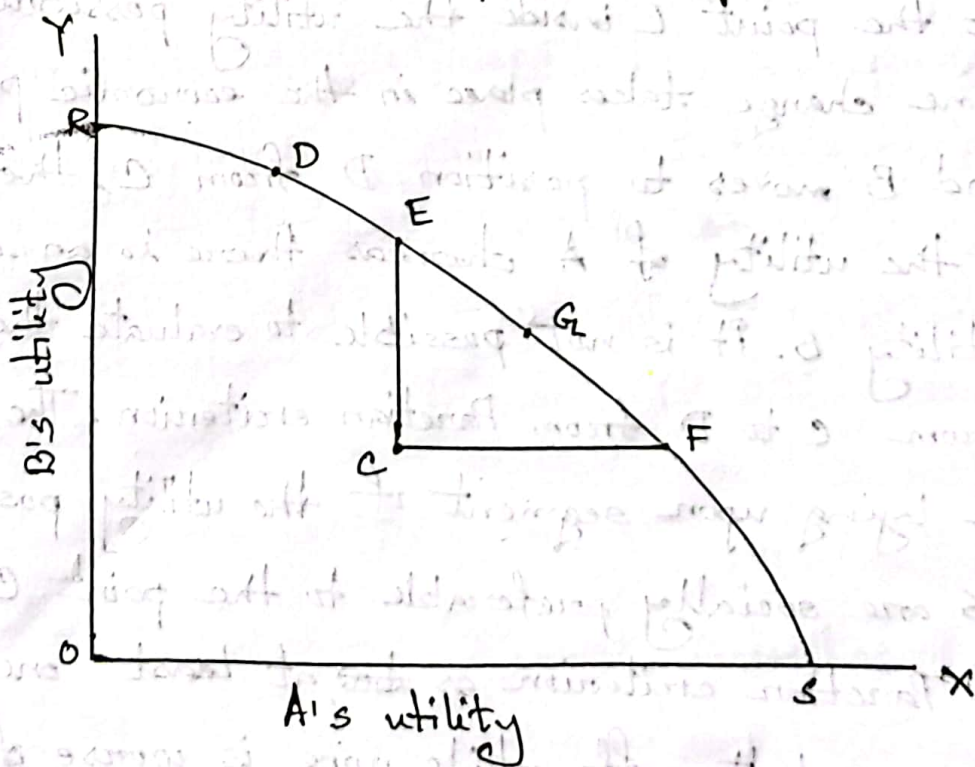
- (ii) There is constancy of the tastes of the individuals.
- (iii) There can be ordinal measurement of utility.
- (iv) The inter-personal comparisons of utilities are possible.

(v) There is an absence of externalities in production and consumption.

Suppose there are two economic situations A and B. The former is socially preferable to B, in case those who gain from A can compensate the losers i.e. they can bribe the losers to accept the situation A and can still be better off than in situation B. For instance, the chemical factory in an area causes suffering to the people in its vicinity but the factory management compensates them and still the total product obtained by society is more than what it would have been, had the factory been closed. There is a net increase in the welfare of the society.

J.R. Hicks put forward his compensation criterion which is the reverse of Kaldor's criterion. According to him, if an economic change makes some people better off and others worse off and the losers cannot bribe the gainers not to make the given change, then the

is an increase in the net welfare of the society from that economic change. Suppose the government proposes to impose a tax which will benefit some and harm others. If the latter fail to bribe the government officials not to introduced the proposed tax, then the given change in tax policy will result in a net increase in the welfare of the society. The compensation principle suggested by Kaldor and Hicks, even though stated in different words, actually means the same thing. The only difference is that Hicks has given his criterion from viewpoints of losers, whereas Kaldor formulated his criterion from viewpoint of gainers.



In the given figure, A's utility is measured along the horizontal scale and B's utility along the vertical scale.

RS is the utility possibility curve. It indicates the different combinations of the utilities obtained by A and B. The movement downwards along the curve RS shows an increase in the utility of A and a fall in the utility of B. On the opposite, an upward movement along RS curve shows a fall in the utility of A and a rise in utility of B. These movements along the utility possibility curve may be on account of redistribution of income.

Suppose the two individuals A and B are initially at the point C inside the utility possibility curve. If some change takes place in the economic policy and A and B moves to position D from C, there is a fall in the utility of A whereas there is an increase in the utility B. It is not possible to evaluate the movement from C to D from Paretian criterion. The points E, F and G lying upon segment of the utility possibility curve RS are socially preferable to the point C on the basis of Paretian criterion as ~~at~~ at least one individual becomes better off while none is worse off at the points E, F and G. As there is movement from C to D, it has to be seen if the individual B (the gainer) can

compensate the individual A (the loser) and still be better off than C.

According to Kaldor-Hicks criterion, as B compensates A, the income redistribution causes the movement from D to E where A's utility remains as before but B is still better off. It signifies that there is a net increase in the welfare of the society just by the redistribution of income. Even the movement from D to G will result in a net increase in social welfare because at G both A and B are better off than at the original position C. According to Kaldor-Hicks criterion, it is not necessary to pay compensation actually to judge whether the social welfare has increased or not.

Only requirement is that the gainer should be potentially able to compensate the loser for the loss ~~of~~ in his welfare and still be better off.

Ques-04:- Define social welfare function. How does it could be used a criteria to measure social welfare? Explain.

Answer:-

Social Welfare Function:-

In welfare function, a social welfare function is a function that ranks social states (alternative complete descriptions of the society) as less desirable, more desirable, or indifferent for every possible pair of social states. The social welfare function provides the government with a simple guideline for achieving the optimal distribution of income.

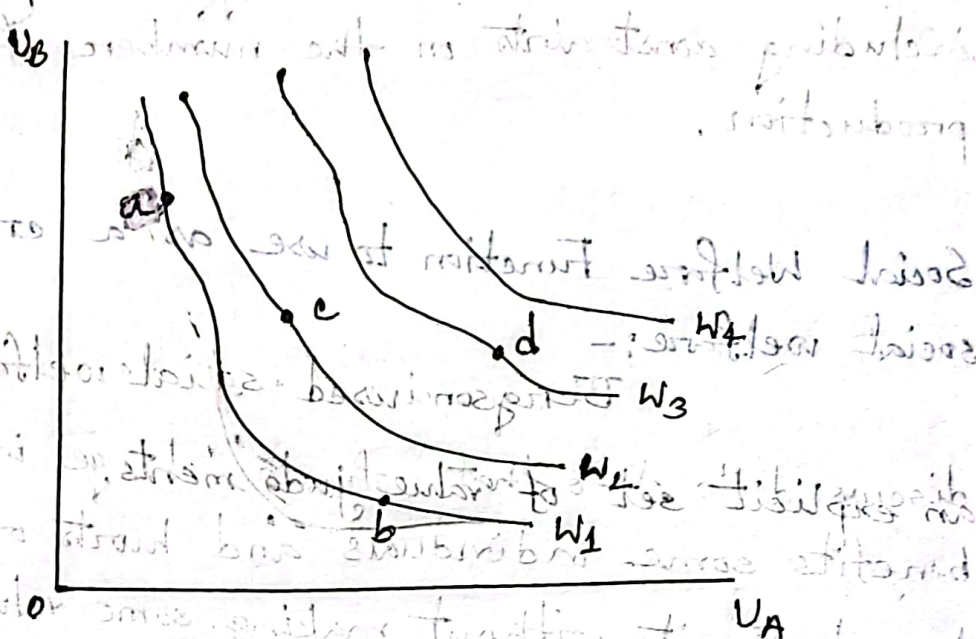
The social welfare function is analogous to the consumer theory of indifference-curve-budget constraint tangency for an individual, except that the social welfare function is a mapping of individual preferences or judgements of everyone in the society as to collective choices, which apply to all, whatever individual preferences are for (variable) constraints or factors of production. One point of a social welfare function is to determine how close the analogy is to an ordinal utility function for an individual with at

least minimal restrictions suggested by welfare economics, including constraints on the numbers of factors of production.

Social Welfare Function to use as a criterion to measure social welfare:-

The various welfare criteria so far discussed to show that when a change in the economy benefits some individuals and hurts others it is impossible to evaluate it without making some value judgement about the deservingness of the different individual or groups. Bergson suggested the use of an explicit set of value judgements in the form of a social welfare function. A social welfare function is analogous to the consumer's utility function. It provides a ranking of alternative states (situations, configurations) in which different individuals enjoy different utility levels. If the economy consists of two individuals, the social welfare function could be presented by a set of social indifference contours (in utility space) like the ones shown in figure. Each curve is the locus of combinations of utilities of A and B which yield the same level of social welfare. The further to the right ~~to~~ a social indifference contour is, the higher the

level of social welfare will be.



With such a set of social indifference contours alternatives states in the economy can be unambiguously evaluated. For example, a change which would move the society from point b to point c (or d) increases the social welfare. A change moving the society from a to b leaves the level of social welfare unaltered. The problem with the social welfare function is that there is no easy way of constructing it. Its existence is axiomatically assumed in welfare economics. Somebody in the economy must undertake the task of comparing the various individuals or groups and rank them according to what he thinks their worthiness is. A democratically elected government could be assumed to make such value judgements which would be acceptable

by the society as a whole. This is implicitly or explicitly assumed when use is made to the apparatus of the social welfare function. It should be noted that the social welfare function cannot be used to derive social (or community) indifference curves in output space (analogous to the indifference curves of a single individual) without taking into account the distribution of income among the various individuals in the economy.

Ques-05:- What are the major assumptions of social welfare maximization in a $2 \times 2 \times 2$ model?

Answer:-

The assumptions of welfare maximization in the simple two-factor, two-commodity, two-consumer ($2 \times 2 \times 2$) model are listed below:-

(i) There are two factors, labor L , and capital K , whose quantities are given (in perfectly inelastic supply). These factors are homogenous and perfectly divisible.

(ii) Two products, X and Y , are produced by two firms. Each firm produces only one commodity. The production functions give rise to smooth isoquants, convex to the origin with constant returns to scale. Indivisibilities in the production processes are ruled out.

(iii) There are two consumers whose preferences are represented by indifference curves, which are continuous, convex to the origin, and do not intersect.

(iv) The goal of consumers is utility maximization and the goal of firms is profit maximization.

(v) The production functions are independent. This rules out joint products and external economies and diseconomies in production.

(vi) The utilities of consumers are independent. Bandwagon, Snob and Veblen effects are ruled out. There are no external economies or diseconomies in consumption.

(vii) The ownership of factors, that is, the distribution of the given L and K between the two consumers, is exogenously determined.

(viii) A social welfare function, $W = f(U_A, U_B)$, exists. This permits a unique preference ordering of all possible states, based on the positions of the two consumers in their own preference maps. This welfare function incorporates an ethical valuation of the relative deservingness and/or worthiness of the two consumers.

Ques-06:- Derive the grand utility possibility frontier and determine the 'point of bliss'.

Answer:-

The Grand Utility Possibility Frontier:-

The social welfare function is one of the tools which we will use in finding the situation which maximizes social welfare. For this purpose, however, we need another tool, the grand utility possibility frontier. This shows the maximum utility attainable by B, given the utility enjoyed by A from any given product-mix.

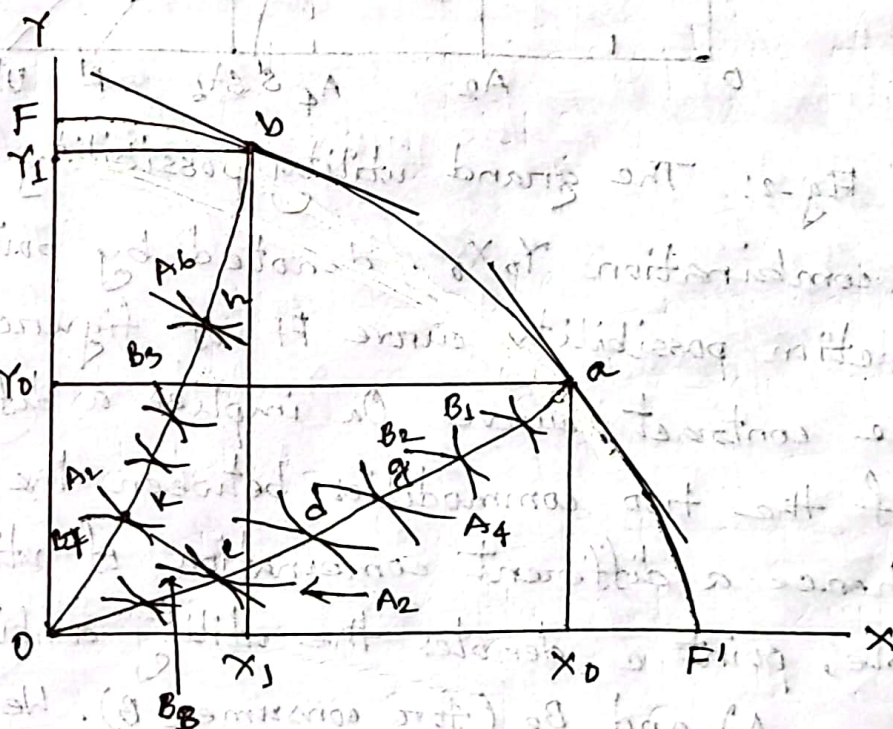


Fig-1

We know each product-mix (combination of X and Y) can be distributed optimally among the two consumers in an infinite number of ways, represented by the points of the Edgeworth contract curve of exchange corresponding

to that particular output combination. Take, for example

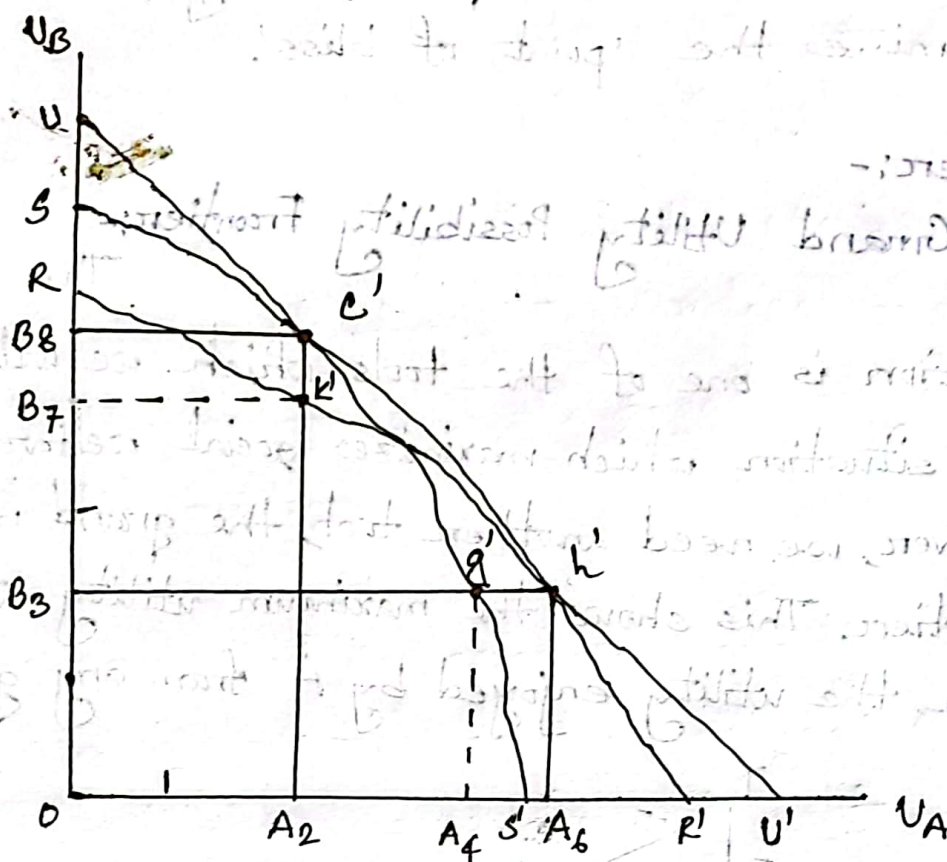


Fig-2: The grand utility possibility frontier
the combination $Y_0 X_0$, denoted by point a on the production possibility curve FF' of figure-1. Each point on the contract curve Oa implies a different distribution of the two commodities between the two consumers, and hence a different combination of utilities. For example, point c denotes the utility combination A_2 (for consumer A) and B_8 (for consumer B). We can plot this utility combination in the utility space. Point c' in fig-2 represents the $A_2 B_8$ utility combination; it shows the maximum utility attainable by B (B_8), given the utility enjoyed by A (A_2) from the distribution of $Y_0 X_0$ as shown by point c in fig-1. We say that c in output space

maps' into point c' in utility space.

We may repeat this process for any other distribution of the given $Y_0 X_0$ product mix. For example, at point g in fig-1 the two consumers enjoy the utilities A_4 and B_3 . This utility combination is shown by point g' in the utility space of fig-2. Point g in output space maps into g' in utility space. Point g' shows the maximum utility attainable to B (B_3) given the utility of A (A_4) from the distribution of $Y_0 X_0$ denoted by point g in fig-1. By mapping all the points of the contract curve Oa into corresponding points in the utility space we obtain the utility possibility frontier for the particular commodity-mix $Y_0 X_0$ (curve ss' in fig-2). The utility possibility curve ss' is drawn for ^{the} specific product-mix $Y_0 X_0$, and it shows maximum utility possibilities when the economy produces this specific combination of commodities. Since there is an infinite number of points on the PPC curve, there must be an infinite number of utility possibility curves. For example, assume that the economy produces the output-mix $Y_1 X_1$ denoted by point b on the PPC curve of fig-1. The points on the Edgeworth contract curve Ob show Pareto-optimal distributions of the product-mix $Y_1 X_1$. Point h shows the utility combination $A_6 B_3$, which is depicted by point h' in the utility space of fig-2. Similarly, point k in output space is mapped into

point k^1 in utility space. The remaining points of the Ob contract curve are mapped into the points of the utility possibility frontier RR' in fig-2.

In summary, each point on the PPC gives rise to a utility possibility frontier. The envelope of these utility possibility frontiers is the grand utility possibility frontier of the economy.

The 'grand' utility associated with this distribution is represented by point h^1 in the utility space of fig-2. Repetition of this procedure for each point on the PPC yields the 'envelope' utility possibility frontier. This is shown by the curve UV' in fig-2. Thus, the grand utility possibility frontier is the locus of utility combinations (of the two consumers) which satisfy the marginal condition $MRPT = MRS$ for each commodity-mix. Each point of the 'envelope' shows the maximum B's utility for any given feasible level of A's utility, and vice versa.

The Point of Bliss:-

In fig-3, the grand utility possibility frontier is combined with the social welfare function shown by the set of social indifference contours.

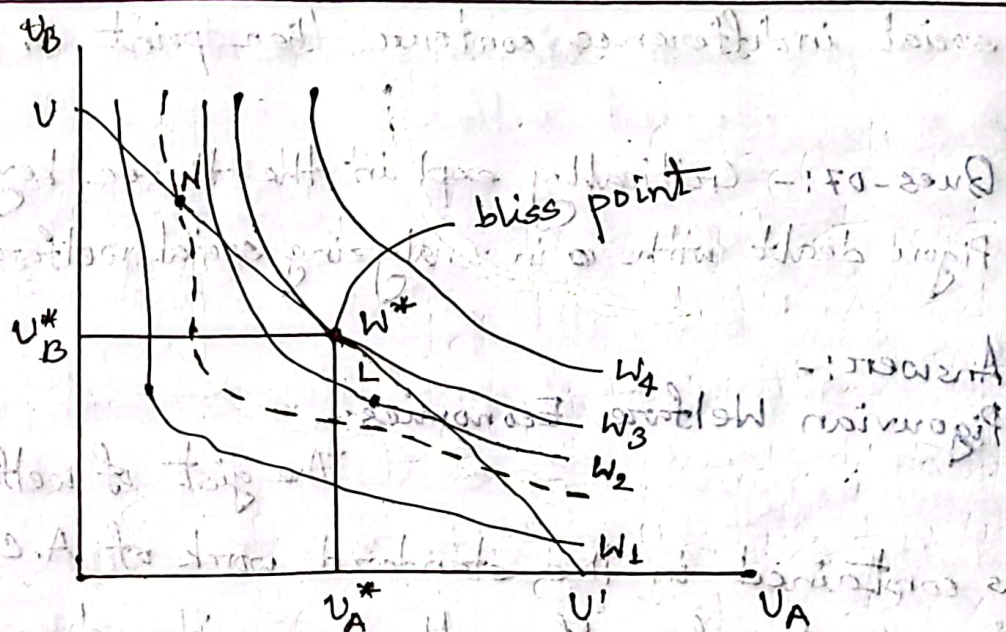


Fig-3:- Maximization of social welfare

Social welfare is maximized at the point of tangency of the 'envelope' utility possibility frontier with the highest possible social indifference contour. This point is called 'the point of bliss'. It is denoted by W^* in fig-3. The maximum social welfare attainable in our example is the level implied by the indifference contour W_3 . The two consumers will enjoy the levels of utility U_A^* and U_B^* .

The point of welfare maximization is a Pareto-optimal state. However, a large number of points below the grand utility frontier, although not Pareto-optimal, yield a higher level of social welfare than points on the utility frontier. For example, point N in fig-3 is a Pareto-optimal situation while point L is not. Yet L lies on a higher

social indifference contour than point N.

Ques-07:- Critically explain the three key things that Pigou dealt with in analyzing social welfare.

Answer:-

Pigouvian Welfare Economics:-

The gist of welfare economics is contained in the standard work of A.C. Pigou, the economist of welfare. He is considered as the father of welfare economics. Pigou dealt with three things:-

- (1) A definition of economic welfare
- (2) Spelling out the condition under which welfare is maximized and
- (3) Pronouncement of policy recommendation for increasing welfare.

Prof. Pigou gave a clear meaning to the concept of welfare. He defined individual welfare as the sum of satisfactions obtained from the use of goods and services. Social welfare is the summation of all individual welfare in a society. Since general welfare is very wide and complicated, he limited his study to economic welfare. He defined economic welfare as that part of social welfare "that can be brought directly

or indirectly into relation with the measuring rod of money."

Pigou regards economic welfare and national income as coordinate. He lays down two conditions for maximization of welfare:

(i) Given the taste and income distribution, an increase in national income represents an increase in welfare,

(ii) For welfare maximization, the distribution of national income is equally important.

If national income remains constant, transfer of income from rich to the poor would improve welfare. With income subject to diminishing marginal utility, transfers of income from the rich to the poor will increase social welfare by satisfying the more intense wants of the poor. Thus it is economic equality that maximizes welfare.

Prof. Pigou had a dual criterion for detecting the increase in social welfare. First, he measured the economic welfare of the society in money value and thus, given the supply of resources, an increase in national dividend meant an increase in social welfare. Second, Pigou favored an income equalization policy and therefore, reorganization of the economy which increases the share of the poor without offsetting adversely "productive effort enterprise and

development of capital equipment was to be taken as a gain in social welfare.

Pigou has made a distinction between private and social costs. The private marginal cost of a commodity is the cost of producing an additional unit. The social marginal cost ~~is~~ is the expense or damage to society as a consequence of producing that commodity. Private marginal benefit can be measured by the selling price of the commodity.

Social marginal benefit refers to the total benefit that society gets from the production of an additional unit. By making a distinction between social and private valuations of economic activity, Pigou paved the way for the analysis of external effects or externalities in social welfare economics.

The presence of external effects in production was seen by Prof. Pigou in the divergence between social net product and private net product. He defined social net product "as the aggregate contribution made to the national dividend" and the private net product as the contribution which is capable of being sold and the proceeds added to the earnings of the person responsible

for investment.

The divergence between the two products shows itself in the form of external effects of production associated with marginal increments of output. In some cases social net product is more than the private product is greater than the social product. As an example of the former, Pigou pointed out to the greater social benefit from technical training of workers by a private firm. As an illustration of the latter, he cited the fact that the smoke rising from the chimneys of private factory spoils the atmosphere of the locality and increase the laundry bills of the people of the neighbourhood. But people are not compensated in any way by the factory owner. He was of the opinion that the state should equalize the private net product with social net product, if in an industry where private net product is more, it should be taxed, and if another industry shows a lesser private net product, it ought to be subsidized. Of course Prof. Pigou recognized that the divergence between private net product and social net product cannot always be quantified and measured in terms of money.

Prof. Pigou made the first attempt to lay down the conditions of social optimum which he termed "the

ideal output" of the economic system as a whole. In his view, the social optimum prevails when marginal social products are equal in all industries and thus production of real wealth is maximized. Assuming that all productive resources are being employed and that there is no cost of movement between different occupations and places, it can be concluded that the national dividend is the largest when the values of the marginal social net products are equal in all industries. If this arrangement prevails, the society is its "ideal output".

Pigovian welfare conditions presuppose the existence of the following assumptions:-

(1) It was assumed that everybody as consumer acts rationally to maximise his satisfaction.

(2) Pigou believed that utility is not measurable cardinally. But interpersonal and intra-personal comparisons of utilities are also possible so that it is possible to find out quantitatively whether welfare has increased or decreased.

(3) Pigou put forward a basic postulate that a man possesses equal capacity for satisfaction when placed in similar circumstances. Pigou held that different people derive the same satisfaction out of the same

real income.

(4) Pigou also believed that the marginal utility of money income falls as money income increases. As a result, the marginal utility of an addition to the income of a poor man is greater as compared to the loss of utility from the loss of the same amount of income to a rich man.

Ques-08:- Does perfect competition ensure welfare maximization? Examine.

Answer:-

The problem of welfare maximization is purely 'technocratic'. We know the bliss point is attained by equalising the slopes of isoquants, the slopes of the indifference curves, and the slope of the production possibility curve to the (equalised) slope of the indifference curves. Thus, welfare-maximizing solution does not depend on prices. The general equilibrium solution reached with perfect competition is the same as the situation implied by the bliss point of the welfare maximization problem.

(a) Profit maximization by the individual firm implies that whatever output the firm may choose as the most profitable must be produced at a minimum cost. Cost minimization is attained if the firm chooses the input combination at which the marginal rate of technical

substitution of the two factors is equal to the input price ratio.

$$MRTS_{L,K} = \frac{w}{r}$$

Since perfect competition all firms are faced by the same set of factor prices, it follows that

$$MRTS_{L,K}^X = MRTS_{L,K}^Y = \frac{w}{r}$$

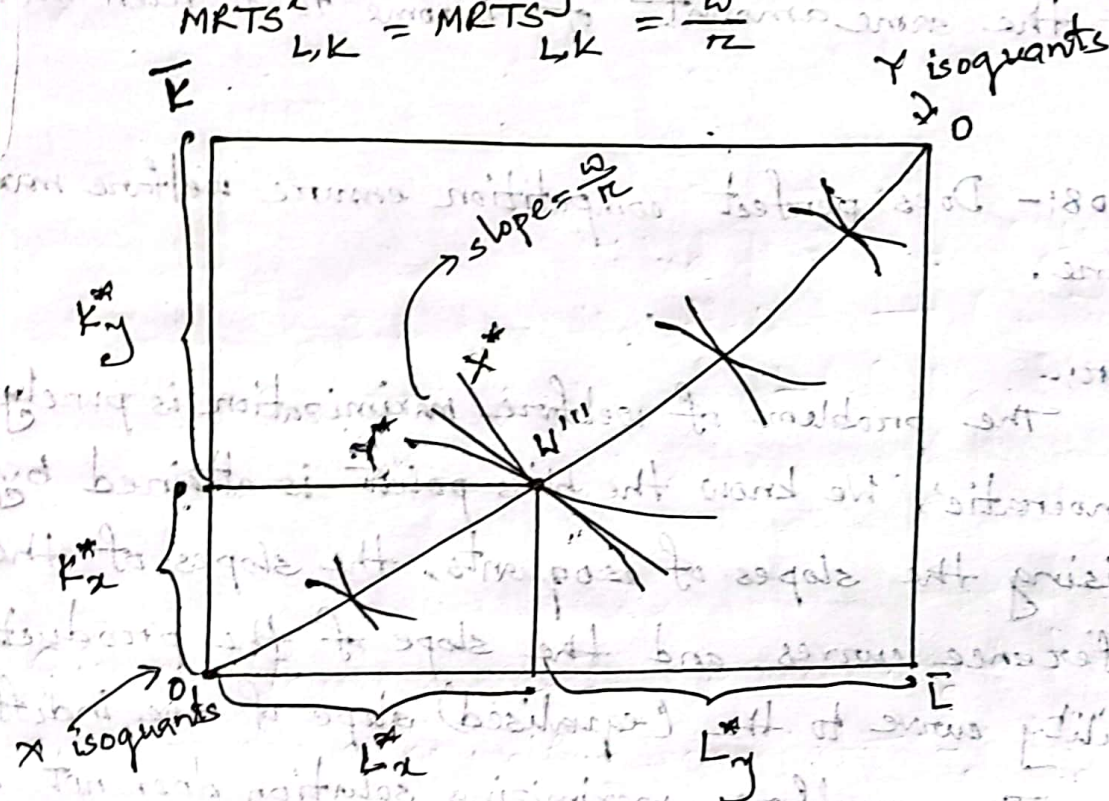


Fig-1

Thus in fig-1 the slope of the contract curve at W''' must be equal to w/r in the perfectly competitive input markets, since W''' is the point of the (equalised) slope of the isoquant Y^* and X^* .

(b) Utility maximization by each individual requires the choice of the product mix where the marginal rate of substitution of the two commodities is equal

to the ratio of their price

$$MRS_{x,y} = \frac{P_x}{P_y}$$

Since in perfect competition all consumers are faced by the same commodity prices, it follows that

$$MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$$

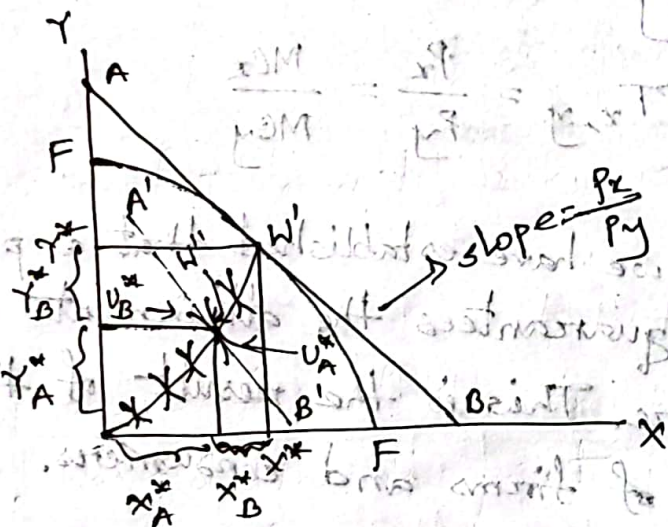


Fig. 2

Thus in fig. 2 the slope of the contract curve at W'' must be equal to the P_x/P_y ratio in perfectly competitive commodity markets, since W'' is the point of the (equalized) slope of the indifference curves of the two consumers.

(c) We have shown that at the bliss point the (equalized) slope of the indifference curves (the 'common' $MRS_{x,y}$) is equal to the slope of the production possibility curve at W'' . Thus at W'' , which defines the product-mix that maximizes social welfare, we have

$$MRPT_{x,y} = \frac{P_x}{P_y}$$

(d). Finally, a firm in a perfectly competitive market maximizes its profit by setting its marginal cost equal to the market price of the commodity. Consequently we have

$$MRPT_{x,y} = \frac{P_x}{P_y} = \frac{MC_x}{MC_y}$$

Thus we have established that a perfectly competitive system guarantees the attainment of maximum social welfare. This is the result of the maximizing behaviour of firms and consumers.

Ques-02:- 'Pareto optimality is a necessary but not sufficient condition for social welfare maximization' - justify the statement considering a 2x2x2 model.

Answer:-

Pareto optimality refers to any change that makes at least one individual better-off and no one worse-off is an improvement in social welfare. Conversely, a change that makes no one better-off and at least one worse-off is a decrease in social welfare.

For the attainment of a Pareto-efficient situation in an economy three marginal conditions must be satisfied: (a) Efficiency of distribution of commodities among consumers (efficiency in exchange); (b) Efficiency of the allocation of factors among firms (efficiency in production); (c) Efficiency in the allocation of factors among commodities (efficiency in the product-mix). We'll satisfy these three conditions to show that the Pareto optimal state is a necessary but not sufficient condition for maximum social welfare below.

(a) Efficiency of distribution of commodities among consumers

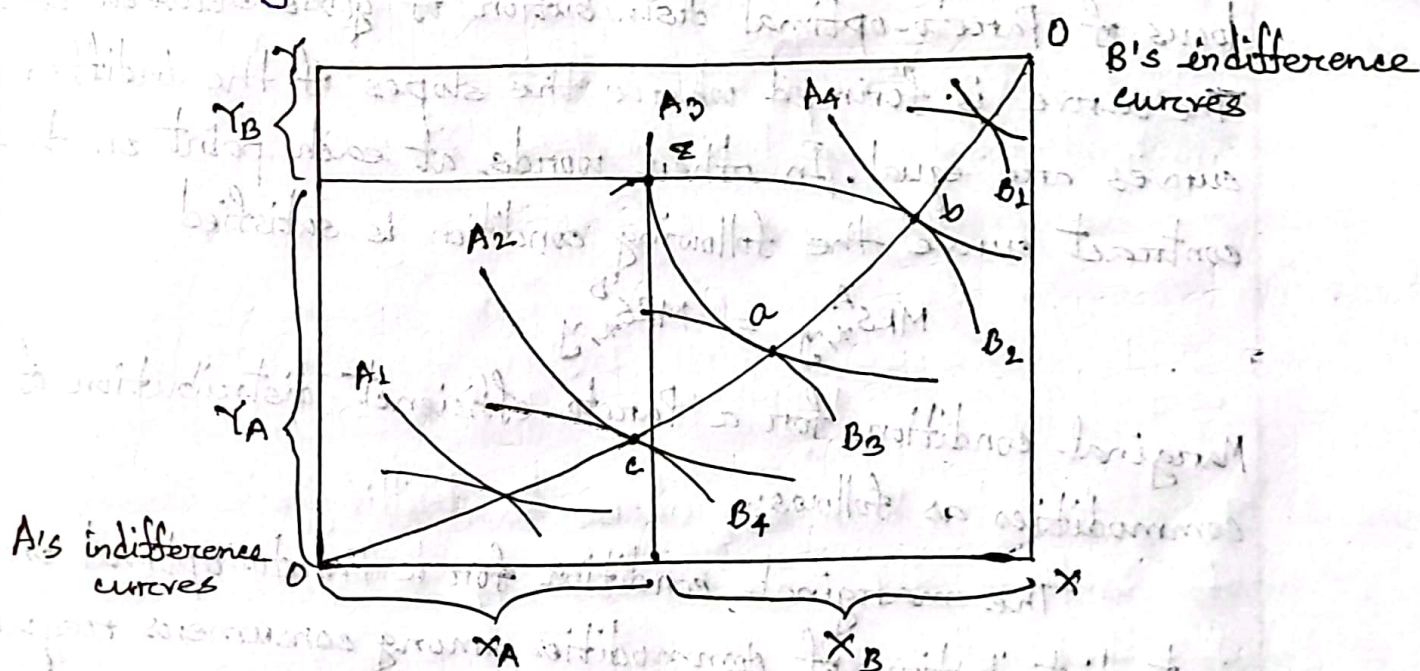


Fig-1:- Edgeworth box of exchange

Applying the Pareto optimality criterion to the case of distribution of commodities Y and X, that is, a distribution of the given commodities X and Y between the two consumers is efficient if it is ~~possible~~ impossible by a redistribution of

these goods to increase the utility of one individual without reducing the utility of the other. In fig-1 we show the Edgeworth box for the given commodities X and Y. In the figure, point z is inefficient since a redistribution of the commodities such as to reach any point between a and b increases the utility of both consumers. A movement to a increases the utility of B without reducing the utility of A. Similarly, the distribution implied by b increases the utility of A without reducing the utility of B. Thus all the points from a to b represent improvements in social welfare compared with the distribution at z. The contract curve shows the locus of Pareto-optimal distribution of goods between consumers. The curve is formed where the slopes of the indifference curves are equal. In other words, at each point on the contract curve the following condition is satisfied

$$MRS_{x,y}^A = MRS_{x,y}^B$$

Marginal condition for a Pareto-efficient distribution of given commodities as follows:-

The marginal condition for a Pareto-optimal or -efficient distribution of commodities among consumers requires that the MRS between two goods be equal for all consumers.

(b) Efficiency of allocation of factors among firm-producers

Allocation of given resources K and L using the Edgeworth box of production is shown in fig 2.

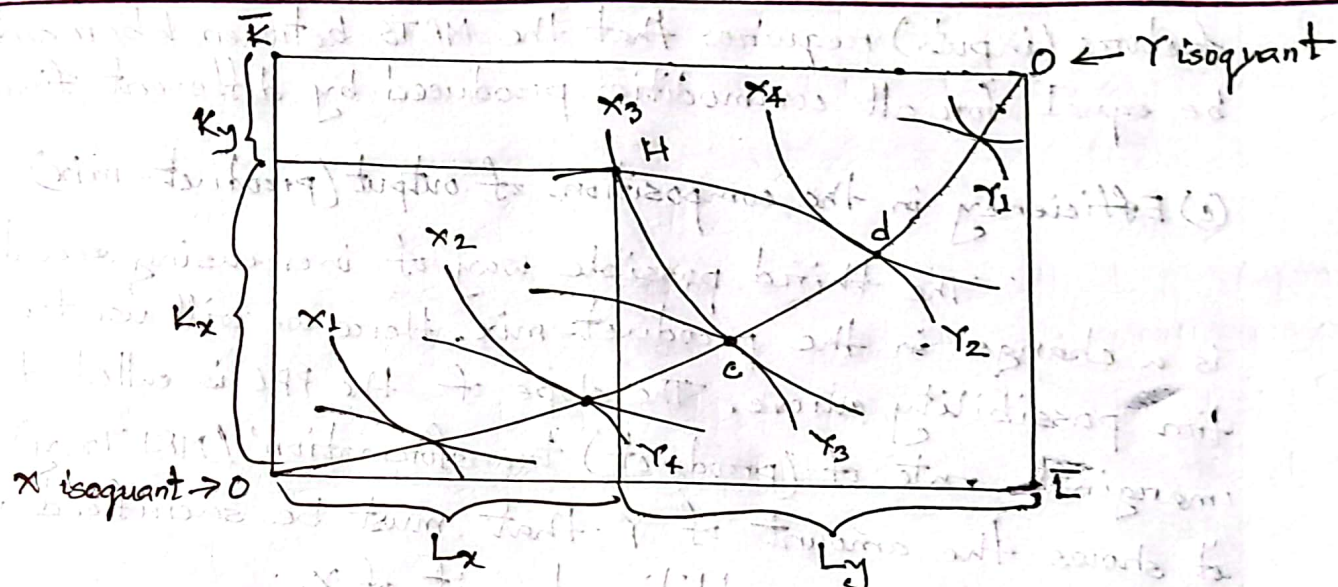


Fig-2:- Edgeworth box of production

Only points on the contract curve of production are Pareto-efficient. Point H is inefficient, since a reallocation of the given K and L between the producers of X and Y such as to reach any point from c to d inclusive results in the increase of at least one commodity without a reduction in the other.

The contract curve is the locus of points of tangency of the isoquants of the two firms which produce X and Y that is, points where the slopes of the isoquants are equal. Thus at each point of the contract curve the following condition holds

$$MRTS_{L,K}^X = MRTS_{L,K}^Y$$

The marginal condition for a Pareto-optimal allocation of factors among firms as follows:-

The marginal condition for a Pareto-optimal allocation of

factors (inputs) requires that the MRTS between labor and capital be equal for all commodities produced by different firms.

(c) Efficiency in the composition of output (product-mix)

The third possible way of increasing social welfare is a change in the product-mix. Here we will use the production possibility curve. The slope of the PPC is called the 'marginal rate of (product) transformation' ($MRPT_{x,y}$) and it shows the amount of Y that must be sacrificed in order to obtain an additional unit of X .

The marginal condition for a Pareto-optimal composition of output requires that the $MRPT$ between any two commodities be equal to the MRS between the same two goods:

$$MRPT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$$

Since the $MRPT$ shows the rate at which a good can be transformed into ^{another} (on the production side), and the MRS shows the rate at which consumers are willing to exchange a good for another, the rates must be equal for a Pareto-optimal situation to be attained. Suppose that these rates are unequal. For example assume

$$MRPT_{x,y} = \frac{2Y}{1X} \quad \text{and} \quad MRS_{x,y} = \frac{1Y}{1X}$$

that is,

$$MRPT_{x,y} > MRS_{x,y}$$

The above inequality shows that the economy can produce two units of Y by sacrificing one unit of X , while the consumers are willing to exchange one unit of Y for one unit of X . Clearly

the economy produces too much of X and too little of Y . Welfare therefore can be increased by increasing the production of Y and decreasing the production of X .

In summary, a Pareto-optimal state in the economy can be attained if the following three marginal conditions are fulfilled:-

- (i) The $MRS_{x,y}$ between any two goods be equal for all consumers.
- (ii) The $MRTS_{L,X}$ between any two inputs be equal in the production of all commodities.
- (iii) The $MRPT_{x,y}$ be equal to the $MRS_{x,y}$ for any two goods.

A situation may be Pareto-optimal without maximizing social welfare. However, welfare maximization is attained only at a situation that is Pareto-optimal. In other words, Pareto optimality is a necessary but not sufficient condition for welfare maximization.